An M-Nary SAR Image Change Detection Based on GAN Architecture Search

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***Abstract*— Change detection (CD) in synthetic aperture radar (SAR) images aims to detect changed areas by considering the changes in backscattering coefficients. However, the changes can be further divided into positive and negative changes in terms of the increase or decrease of backscattering coefficient, so the CD task can be divided into binary and ternary according to the number of existent categories. This article introduces an M-nary (binary or ternary) SAR CD procedure based on the generative adversarial network (GAN) and neural architecture search (NAS) strategy to detect which changes exist in the SAR image pair and design specialized classifiers for both binary CD (BCD) and ternary CD (TCD). First, a difference image (DI) generation approach based on the salient changed region extraction and neighborhood information is designed for a robust difference representation on the M-nary CD. Due to the further subdivision of changes, the insufficiency of labeled data presents the M-nary CD with a dilemma. Concerning the lack of labeled information, this article presents a labeled sample generation strategy based on the GAN architecture search to supplement sample data. Since GAN training is inherently unstable, NAS provides an effective means of searching GAN architecture automatically and ameliorates the reliability of generated samples. During the architecture search procedure, a double-phase evolutionary search strategy is introduced to further improve the stability of GAN training. The experimental results with theoretical analysis prove the validity, robustness, and potential of our method in synthetic as well as real SAR datasets.**

***Index Terms*— Change detection (CD), difference image (DI), generative adversarial network (GAN), neural architecture search (NAS), synthetic aperture radar (SAR) image.**

I. Introduction

IMAGE change detection (CD) aims to detect regions of change in temporal images of the same scene [1], which has been widely applied in a great number of fields in diverse disciplines, such as land resources management, agriculture and forestry monitoring, natural disaster monitoring with assessment, and so on [2], [3], [4], [5]. As an active microwave remote sensor imaged by pulse compression and synthetic aperture technique, synthetic aperture radar (SAR) has been an effective medium for CD with its characteristics of all-weather, all-time, strong penetrability, strong detection ability to geographical texture features, as well as multiband and multipolarization scattering [6].

Early literature (see [7], [8], [9]) shows the basic process of the SAR image CD, where difference image (DI) generation and DI analysis are the focus of SAR image CD procedure. The purpose of generating DI is to describe the change information of the image pair. Because of the unique SAR imaging mechanism, the DI of SAR images requires to take the speckle noise suppression into consideration. Based on ratio operator, the logarithmic ratio (LR) operator [10] and the mean ratio (MR) [11] operator are two widely used operators. Besides, there are also several popular DI generation algorithms, such as neighborhood-based ratio (NR) operator [9] and Wavelet fusion [12], which are devoted to learning the difference representation of remote sensing (RS) image pair. The DI analysis can be seen as a process of image classification. Threshold and clustering are two common strategies for DI analysis. Apart from the classical Kittler and Illingworth (KI) [13] method, there have been several threshold methods, such as generalized KI (GKI) [7] threshold algorithm, histogram optimization-based threshold [14] method, local fitting and semiexpectation maximization [15] algorithm, and so on. Most of the threshold methods require to build a model to fit the class condition distribution of changed pixels and unchanged pixels at first. Compared with threshold, clustering methods, such as fuzzy C-means (FCM) [16], fuzzy local information C-means (FLICM) [17], and reformulated fuzzy local-information C-means (RFLICM) [18], are more flexible without modeling any data distribution, but the detection performance is still limited by shallow difference feature.

Nowadays, according to whether prior knowledge is needed or not, CD techniques can also be categorized as supervised and unsupervised approaches. Without the external supervision ground reference data, unsupervised methods rely on basic principles injected in algorithms or the distribution of values in DI. Several unsupervised methods work to obtain more accurate difference information, such as the improved NR (INR) method [19], graph-based data fusion (GBF) method [20], and mean shift enforced information theoretic CD (MS-ITCD) [21] method. In addition, there are also some unsupervised methods combined with several advanced techniques, such

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as Markov random field (MRF) [22], principal component analysis (PCA) [6], evolutionary algorithm (EA) [23], saliency extraction [24], sparse representations [25], deep neural networks (DNNs) [26], and so on. However, subtle and accurate detection results in unsupervised methods are always hard to obtain due to the lack of priori knowledge. Supervised methods have the capability of learning from draw data, which utilize the prediction model exploited with training samples to achieve the accurate change map (CM) [27], [28], [29]. With the development of the DNNs, researchers have applied various deep models to the SAR image CD task. Relying on the powerful abilities of nonlinear transformation, deep feature extraction, and automatic learning, CD methods applied DNN models, such as convolutional neural network (CNN) [2], generative adversarial network (GAN) [30], recurrent neural network (RNN) [31], graph convolutional network (GCN) [32], and transformer [33], have achieved great results [34], [35]. Nevertheless, most DNN models require a large amount of labeled data for training. Although there are already several methods based on pre-detection to produce relatively high-quality pseudo-labels, these methods require manual design for model architectures, which is time-consuming, laborious, and parameter-sensitive. Besides, these pre-detection methods are difficult to balance both the authenticity and diversity of generated pseudo-labels at the same time. Moreover, the quality of labeled data obtained by predetection has a great impact on the performance of trained classifier.

Most CD methods focus on achieving the binary CM, which only aim to detect where change has been incurred or not. In fact, the CD task includes not only detecting the variation location but also distinguishing the specific types of changes that occur there [26]. For multispectral images, there has been a lot of research on multiclassification [1], [36], [37], which distinguishes different various change classes of change by analyzing abundant spectral feature changes as well as semantic information changes. Unlike multispectral images with abundant spectral features, SAR actively launches the microwave signals and forms images by receiving the reflected signals. Different objects information can be reflected by different backscattering values of SAR sensor. Therefore, it is possible to discriminate the specific changes according to the increase or decrease of the corresponding backscattering values, where the changes can be classified into positive and negative changes. In practice, the positive changes often exhibit as land-cover transitions (such as buildings rebuilt and lake reclamation), while the negative changes exhibit as disaster loss (such as wildfire or flood). There also has been a few research on ternary SAR image CD. Bazi et al. [8] first proposed the concept of TCD and determined the number of decision thresholds corresponding to typologies of changes, which considered the circumstance that changes present on both sides of the histogram of DI. Based on MRF, Yousif and Ban [4] proposed a TCD approach with the nonlocal means (NLMs), which modeled the local interaction between pixel-level labels, and considered the nonlocal interaction between pixel-level labels. Besides, deep learning is also applied in TCD to seize the critical recognition information in SAR images. For instance, Su et al. [38] presented a deep learning and mapping (DLM) model oriented to the TCD task. In [39], we took advantages of sparse autoencoder (SAE) and CNNs to solve TCD problem by learning more robust change features. However, there are still some problems that limit the performance of these TCD methods. On the one hand, since DI has a great influence on detection performance, the ternary DI aims to distinguish specific change types, which is more susceptible to the inherent speckle noise of SAR images and possesses more visual feature confusion than the binary DI [40]. On the other hand, since TCD divides the changed class into positive and negative changes, it is obvious that the area proportion of each change type in the ground-truth map is unbalanced where a kind of change may be only concentrated in small localized areas. The lack of training samples leads to the difficulty of mining each changed class and thoroughly training the DNNs.

To address the above issues, this article presents a novel GAN architecture search-based M-nary CD (GASCD) framework to identify and distinguish the specific changed types in the SAR image pair. First, an M-nary DI generation strategy based on the salient changed region extraction with neighbor information is proposed to weaken the influence of speckle noise and visual feature confusion. Since the M-nary CD distinguishes the changes into positive change and negative change, the unbalanced proportion between classes results in a scarcity of labeled data for each class, which is required for training the deep learning model. To deal with this problem, this article proposes a label sample generation strategy based on GAN. Considered about the instability and collapse of GAN training, an evolutionary GAN architecture search technique is presented to improve the reliability of generated label samples, where the search of the generator and discriminator is decoupled into double phases with weight-resets strategy to improve the stability of GAN training. The final discovered network GASNet is fully trained to generate reliable samples. Finally, with respect to the BCD or TCD task, the convolution-based classifiers are applied to achieve the final ternary or binary CMs. The main contributions of the proposed approach lie in the following three aspects.

1) We introduce an M-nary CD framework for SAR images, which detects the number of changed types and designs specialized classifiers for both BCD and TCD, respectively.
2) We design a novel DI generation strategy for M-nary CD task, which utilizes the salient changed region extraction with neighbor information to restrain unchanged information and highlight each type of changed information.
3) We propose a labeled sample generation strategy based on GAN architecture search to supplement training data. The architecture search strategy could automatically find superior GAN models in an evolutionary way while ensuring the reliability and accuracy of the generated labeled samples. During the search process, we decouple the search of generator and discriminator into double phases and apply weight-resets method to alleviate the instability of GAN training.

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The rest of this article is organized as follows. The motivation and some background theories are introduced in Section II. The proposed M-nary CD framework is described in detail in Section III. The experimental setting is presented in Section IV. Section V presents the experimental results on simulated and real datasets. Finally, the conclusion of this article is drawn in Section VI.

II. Background

A. Binary and Ternary SAR Image CD

For the two coregistered SAR images I_1 and I_2 acquired at different times t_1 and t_2 over the same geographical area, the BCD task aims to detect the exact changed location of change between the acquired image pair. Therefore, the DI for BCD only takes the change intensity into consideration. As mentioned above, LR operator and MR operator are two well-known techniques for producing DI. The DI of LR can be given by

$$\mathrm{DI}_{\mathrm{LR}} = \left|\log \frac{I_1}{I_2}\right|. \tag{1}$$

The DI of MR for BCD can be obtained as follows:

$$\mathrm{DI}_{\mathrm{MR}} = 1 - \min\left(\frac{\mu_1}{\mu_2}, \frac{\mu_2}{\mu_1}\right) \tag{2}$$

where μ_1 and μ_2 represent the local mean values of image I_1 and I_2, respectively.

Different with BCD, TCD task not only detects the changed regions but also distinguishes the type of changes. Since the image intensity of SAR is related to the underlying backscattering coefficient by the multiplicative model, the changes can be dived into positive and negative changes in terms of increase or decrease in the backscattering values. For TCD, Liang et al. [41] improved the LR and MR operators to distinguish the specific changed type. The DI of LR for TCD can be given by

$$\mathrm{DI}_{\mathrm{LR}} = \log \frac{I_1}{I_2} \tag{3}$$

which removes the absolute value symbol of (1). The DI of MR for TCD can be rewritten as follows:

$$\mathrm{DI}_{\mathrm{MR}} = \begin{cases} 1 - \min\left(\dfrac{\mu_2}{\mu_1}, \dfrac{\mu_1}{\mu_2}\right), & \text{if } \mu_2 > \mu_1 \\ \min\left(\dfrac{\mu_2}{\mu_1}, \dfrac{\mu_1}{\mu_2}\right) - 1, & \text{if } \mu_2 \leqslant \mu_1. \end{cases} \tag{4}$$

By comparing μ_1 and μ_2, changed pixels are classified into the positive and negative changed classes.

B. GAN Architecture Search

GAN is first proposed in [42] and has attracted a widespread attention due to their impressive performance for generating realistic images. A GAN consists of a generator (G) that generates credible new data and a discriminator (D) that distinguishes fake data of G from real data. During the training process, D is trained to maximize the probability that both the real data and generator's data are set as true labels, while G is trained for generating data similar to the real data to fool D. G and D are taken as two competitors in the minimax game as follows:

$$\min_G \max_D V(G, D) = \mathbb{E}_{x\sim p_{\mathrm{data}}}\left[\log D(x)\right] + \mathbb{E}_{z\sim p_z}\left[\log(1 - D(G(z)))\right] \tag{5}$$

where x and z denote the real data and additional noises, respectively, p_{data} represents the real data distribution, and p_z is a prior distribution. $\mathbb{E}$ is the empirical estimate of the probability expectation. However, the optimization of GAN is a minimax game problem, while the generator and discriminator training is difficult to reach Nash equilibrium. Besides, the hyperparameter sensibility and mode collapse incur the instability of the minimax optimization in GAN training. In many works, architecture enhancements have been proven effective to improve GAN performance.

Neural architecture search (NAS) has been widely studied due to the ability of automatically searching the best network architecture so as to greatly reduces human labor. The process of NAS can be described as a bi-level optimization problem

$$\begin{aligned} N^* &= \operatorname*{argmin}_N \ L_v(N|w^*) \\ w^* &= \operatorname*{argmin}_w \ L_t(w|N^*) \end{aligned} \tag{6}$$

where N and w denote the architecture and weight of the network, and L_v and L_t represent the loss function for validation and training. NAS aims to automatically search for the best performing architecture that satisfies certain demands under the condition of the optimal network weight on the training set. Shi et al. [43] have ever applied the NAS technique for searching for better architectures of a fully connected (FC) neural network to solve SAR images CD problems. Recently, there have been several studies on designing NAS techniques to search GANs. AutoGAN [44] first introduced the NAS method to GANs, which applied the reinforcement learning (RL) strategy to search GAN architecture. Automated design of GAN (AGAN) [45] and end-to-end GAN (E2GAN) [46] are also RL-based NAS methods for discovering GAN. However, most works only searches the architecture of G. Besides, AlphaGAN [47] and AdversarialNAS [48] are two gradient-based NAS methods, which employed softmax layer to relax the discrete search space. Different with the previous methods, AdversarialNAS could search G as well as D simultaneously in a differentiable way, but the highly coupled architectures of G and D lead to the increase of negative impact on inferior discriminators. In addition, there are also several EA-based methods, such as evolutionary GAN (EGAN) [49] and coevolution of GAN (COEGAN) [50], which generate and evolve a large population of GAN architectures to achieve the Pareto-front architectures.

C. Motivation

Since SAR is imaged by receiving the reflected signals, the changes regions can be further classified into positive and negative changes according to the increase or decrease of backscattering values. In order to determine the number and category of change types in the image pair, we plan to design an M-nary (binary and ternary) SAR image CD framework,

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which allows the automatic detection of the changed type numbers (one or two) by analyzing the decision thresholds of the histogram for DI and distinguishes the category (positive or negative) of changes by analyzing the location of threshold point in the histogram. According to the number of change types, we plan to design different classifiers for BCD and TCD, respectively.

As posted before, DI has a great influence on CD performance. Different with BCD task, the DI for TCD task requires to further distinguish the specific change type, which is more susceptible to the inherent speckle noise of SAR images and possesses more visual feature confusion than the binary DI. Therefore, we plan to design a robust DI generation algorithm for TCD, which has the ability to highlight each sort changed information and restrain irrelevant information.

Moreover, since TCD requires to distinguish positive and negative changes, the area proportion of each change type in the ground-truth map is unbalanced, which leads to the insufficiency of priori knowledge for detecting each change type. To solve this problem, this article is devoted to designing a sample generation strategy, which can achieve sufficient labeled samples for model training. GANs are competent for general image generation task, but the instability and model collapse result in a quality uncertainty of generated samples. To improve the reliability of generated labeled samples of each type, we consider about a GAN architecture search method to discover a GAN architecture with the best accuracy.

The existing popular GAN architectures are designed by human experts through laborious trial and error testing, and the instability problem in GAN training greatly increases the difficulty of architecture design [51]. Besides, the instability of GAN training leads to the hyperparameter sensibility, mode collapse, as well as gradient vanishing. To alleviate these conditions, one idea is to search G and D simultaneously in a differentiable fashion, but it contributes to the highly coupled architectures of G and D, which potentially increases the negative impact of poor D and hinders discovering the optimal GANs. Therefore, we aim to improve the existent GAN architecture search method to eliminate the adverse effects of high coupling of G and D searching process while improving the stability of GAN training.

III. Methodology

In this section, the proposed framework GASCD will be elaborated for M-nary CD task. The flowchart of GASCD is shown in Fig. 1. First, a ratio operator based on the salient changed region extraction with neighbor information is presented to produce a robust DI for M-nary CD task. To solve the insufficiency of labeled training samples, a labeled data generation strategy based on the GAN architecture search is utilized to discover the best GAN model GASNet, and then, GASNet is applied to supplement training samples with high reliability and good diversity. During the NAS, GASCD decouples the search of generator and discriminator into double phases to improve the stability of GAN training. Finally, the original labeled data and generated labeled data are applied for the soft classifier to achieve the final ternary or binary CM.

A. Difference Information Representation With SNR

A robust DI can extract the efficient difference information representation. The formula of LR as well as MR operator for TCD has been performed as (3) and (4). However, the DIs produced by LR and MR on TCD task are still susceptible to the inherent speckle noise of SAR images and exist more visual feature confusion due to the inclusion of category information. To reduce the influence of speckle noise, we consider about extracting the salient changed region to highlight the changed information of relevant pixels and combining the spatial information of neighbor pixels to restrain irrelevant information. The proposed salient changed and NR (SNR) operator can be shown in Algorithm 1.

Algorithm 1 Salient Changed and NR (SNR) Operator

Input: Two co-registered images I_1 and I_2.
Output: Robust difference image DI_{tcd} for M-nary CD task.

1: **I. Salient change analysis**.
2: Generate an initial DI using LR for BCD task.
3: Segment the initial BCD DI on multi-scale.
4: Calculate the dissimilarity of each patch $\boldsymbol{p}_j^k$.
5: Calculate the salient value of patch $\boldsymbol{p}_j^k$ with other neighbor patches.
6: Combine saliency maps of k levels and evaluate the saliency s at pixel i.
7: **II.M-nary difference representation with neighborhood information**.
8: Calculate LR operator of each pixel on TCD task.
9: Calculate the log ratio of pixel's neighborhood information on TCD task.
10: Compute the weighted sum of two logarithmic ratio operators with salient changed factor s.
11: **return** The robust difference image DI_{tcd} for M-nary CD task.

Both positive and negative changes have greatly changed on the backscattering coefficient values of the original image. Compared with the unchanged regions, the salient changed regions contain more efficient difference information no matter which type of changes. Since context-aware saliency detection [52] could extract attractive and compact saliency regions by calculating the saliency value of each pixel in multiple scales, we develop a context-aware saliency model to detect the salient changed regions of temporal SAR images. As shown in Algorithm 1, the initial DI on BCD task $\mathrm{DI}_{\mathrm{bcd}}$ can be obtained by (1) to guide the detection. Divide $\mathrm{DI}_{\mathrm{bcd}}$ into K levels; then, each level is segmented into patches with different scale sizes. Pixel i can be regarded as salient when the context of the patch $\boldsymbol{p}_i$ centered on i is distinctive compared with other patches. Assuming $\boldsymbol{p}_i^k$ and $\boldsymbol{p}_j^k$ are two vectorized patches corresponding to pixels i and j, the dissimilarity D_{ij}^k on the kth level can be expressed as the distance from the pixel position to the patch center, which can be defined as follows:

$$D_{ij}^k = \frac{\left\| \boldsymbol{p}_j^k - \boldsymbol{p}_i^k \right\|}{\alpha \cdot d\left(\boldsymbol{p}_i^k, \boldsymbol{p}_j^k\right) + 1} \tag{7}$$

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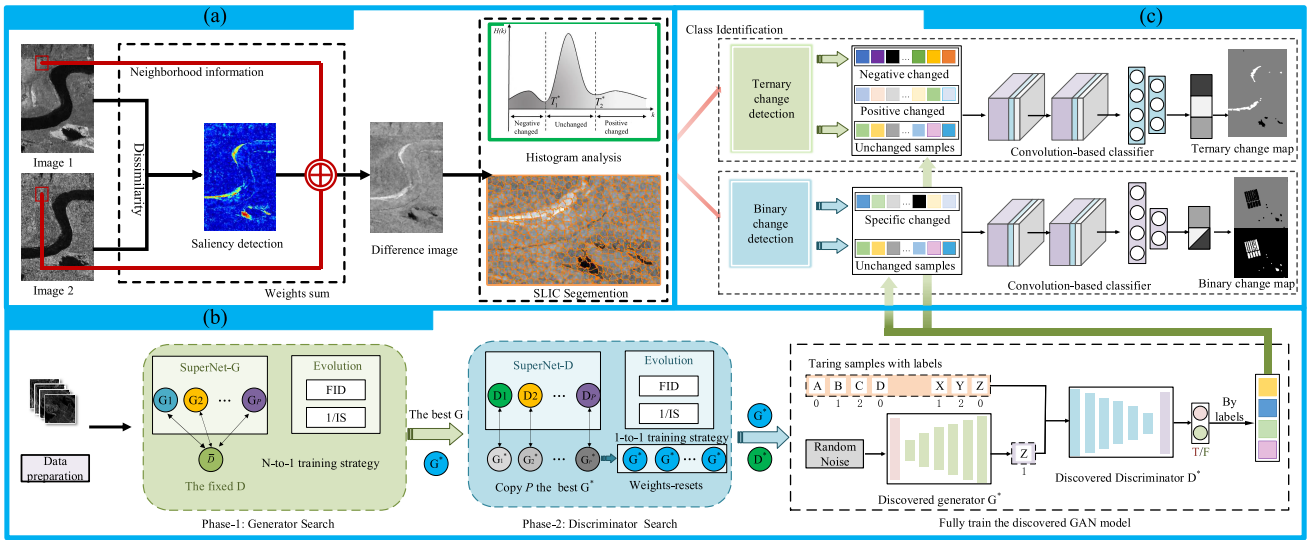

Fig. 1. Flowchart of the proposed method for M-nary SAR image CD. (a) Difference information representation with SNR operator. (b) Labeled sample generation based on GAN architecture search. (c) Convolution-based BCD and TCD.

where α is the adjustment coefficient and $d(\boldsymbol{p}_i^k, \boldsymbol{p}_j^k)$ is the Euclidean distance between patch $\boldsymbol{p}_i^k$ and $\boldsymbol{p}_j^k$. The salient value of each pixel can be evaluated via the dissimilarity D_{ij}^k.

For a patch ${p_i}^k$ on the kth level, all the neighbor patches on the adjacent hierarchy are considered as candidates. Among all these patches, the n most similar patches according to (7) can be obtained and employed to compute the saliency. Thus, the salient value of ${\boldsymbol{p}_i}^k$ can be written as follows:

$$S_i^k = 1 - \exp\left\{-\frac{1}{n}\sum_{m=1}^{n} D_{ij}^k\right\}. \tag{8}$$

The saliency at pixel i is taken as the mean of its saliency at k levels. Combining saliency maps of k levels and interpolating them to the original image size, the saliency of i can be defined as follows:

$$\overline{S_i} = \frac{1}{K}\sum_{k=1}^{K}\left[S_i^k\right] \tag{9}$$

where $[S_i^k]$ indicates the interpolation operation for adjusting the size back to the original image size. The required salient changed factor s can be obtained by normalization operator.

In the salient changed areas, much effective changed information is hoped to be highlighted. But, in the less changed regions, the neighborhood information is considered to reduce speckle noise. Therefore, the SNR operator can be defined as follows:

$$\begin{aligned}\mathrm{DI}_{\mathrm{tcd}}(i,j) = {} & s(i,j)\cdot\log\frac{I_2(i,j)}{I_1(i,j)} \\ & + (1-s(i,j))\cdot\log\frac{\sum_{(x,y)\in\Omega_{ij}} I_2(x,y)}{\sum_{(x,y)\in\Omega_{ij}} I_1(x,y)}\end{aligned} \tag{10}$$

where $\mathrm{DI}_{\mathrm{tcd}}(i,j)$ refers to the difference representation of pixel (i,j) on the DI for M-nary CD and Ω_{ij} denotes the neighborhood of pixel. The value of $s(i,j)$ represents the salient change intensity at (i,j). With respect to the regions with high $s(i,j)$, the difference representation is dominated by each pixel, while for the regions with low $s(i,j)$, the difference of $s(i,j)$'s neighborhood dominates $\mathrm{DI}_{\mathrm{tcd}}(i,j)$ to restrain irrelevant speckle noise.

B. Labeled Sample Generation Based on GAN Architecture Search

Deep learning models execute CD tasks well when there are relatively abundant labeled training samples. However, the quantity of available labels belonging to positive and negative changes is far less than the required sample size. A labeled sample generation strategy is considered to get out of the training sample dilemma. GANs can be applied to produce the new training samples with the advantage of generating new information. But, the training instability of the general GAN limits the reliability of generated samples, which has a serious influence on the final CD performance. Besides, the artificially designed GAN architectures not only require a lot of labors but also hardly find the model with the best performance. Therefore, the NAS technique specially tailored for GANs is considered to execute the conditional image sample generation task for generating positive changed, negative changed, and unchanged labeled samples by category condition.

1) Search Space for GASNet: GAN architecture search can be interpreted as searching a set of cells, including upsampled cell and downsampled cell, and then, use these cells to build a complete GAN. Since convolutional structure possesses a natural hierarchical characteristic, we combine convolutional layers to search for cells with different architectures. To search the architecture of generator (G) and discriminator (D), AdversarialNAS [48] provides a mature thought containing totally 10^{38} different network architectures for GANs. Three upsampled cells are stacked as a generator while four downsampled cells as a discriminator. The search space is shown in Fig. 2.

Directed acyclic graph (DAG) is applied to represent cells, which comprises an ordered sequence of N nodes. SuperNet-G is the alternative architecture in the search space of generator, which contains one FC layer and three stacked upsampled

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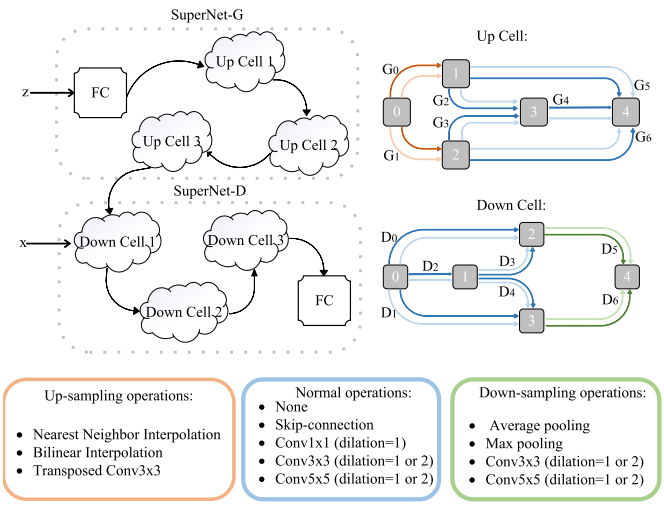

Fig. 2. Search space for GAN architecture search.

cells. Apart from the output of the previous cell Node 0, each cell contains four (1–4) ordered nodes, and each node can select an operation from the candidate set to obtain its previous node through the search algorithm. There are several candidate operations between each two nodes, where each operation is shown as an edge, and only one operation could be selected (the darker edge). As shown in Fig. 2, the edges G_0 and G_1 denote upsampling operators, while other edges in up cells denote normal operators. Through one-hot sequence, each edge can be encoded to select the corresponding operation.

To search for the generator architecture, down cell is formed by the inverted up cell for maintaining their balance, and the SuperNet-D consists of three stacked downsampled cells and a bottom FC layer. In down cells, the edges D_5 and D_6 denote downsampling operators, while other edges indicate normal operations. Based on the above search space settings, the architecture search of $G\&D$ is encoded into searching a collection of one-hot sequences.

2) Evolutionary Architecture Search in Double Phases: The current GAN architecture search methods either only search the single G or search G and D simultaneously, where the former seriously limits the performance of the final GAN model, while the latter contributes to a highly coupled architectures of G and D, which potentially increases the adverse impact of poor D and hinders discovering the best GAN. To alleviate these problems, we decouple the search of G and D into double phases, and each phase consists two steps: weights training and architecture evolution. Since the NAS-based GAN model is applied here to generate the training samples of positive changed labels, negative changed labels, and unchanged labels, we expect to enhance the diversity of generated samples while ensuring the reliability of generated images. Thus, we select inception score (IS) and Fréchet inception distance (FID) as two optimization objectives, where IS describes the diversity of generated pictures, while FID is a method to measure the similarity between image datasets. Therefore, an evolutionary multiobjective optimization algorithm is employed to search a set of Pareto solutions between IS and FID.

Algorithm 2 Evolutionary GAN Architecture Search in Double Phases

Input: SuperNet-G $\mathcal{S}_G$, SuperNet-D $\mathcal{S}_D$, Fitness functions FID and $\frac{1}{IS}$, Population size P, Maximum number of evolution rounds T, Iterations of GAN training I.

Output: The best performing GAN network GASNet with G^* and D^*.

I. Generator architecture search.
$\bar{D} \sim (\bar{m}, \omega_{\bar{D}}) \leftarrow$ Initialize a fixed discriminator $\bar{D}$ with architecture $\bar{m}$ and weights $\omega_{\bar{D}}$.
$\mathcal{P}_G^2 = \{G_1^1, G_2^1, \ldots, G_P^1\} \leftarrow$ Activate all candidate generators in search space uniformly to form the initial population.
$(G_p^1, \bar{D}), p \in \{1, 2, \ldots, P\} \leftarrow$ Initialize all the candidate generators to share the same fixed discriminator as an $N-to-1$ training way.
for $t = 1$ to T **do**
  for $i = 1$ to I **do**
    Sample batch in training set and introduce noise data.
    Train each candidate network G_p^t in $\mathcal{P}_G^t$.
    Update weights $\omega_{\bar{D}}$ and $\omega_{G_i}^t$.
  end for
end for
Evaluate FID and $\frac{1}{IS}$.
$\mathcal{P}'^t_G \leftarrow$ Sort and select the non-dominating individuals on validation set.
$\mathcal{P}_G^t \leftarrow$ Crossover and mutation operator.
II. Discriminator architecture search.
$G \sim (n^*, \omega_{G^*}) \leftarrow$ Copy the optimal generator G^* with n^* and ω_{G^*} discovered in Phase-1.
$\mathcal{P}_D^2 = \{D_1^1, D_2^1, \ldots, D_P^1\} \leftarrow$ Activate all candidate discriminators in search space uniformly to form the initial population.
$(G_p, D_p^1), p \in \{1, 2, \ldots, P\} \leftarrow$ Copy the best generator to pair each candidate discriminator.
for $t = 1$ to T **do**
  for $i = 1$ to I **do**
    Sample batch in training set and introduce noise data.
    Train each candidate GAN (G_p, D_p^t) in P GANs.
    Update weights $\omega_{D_i}^t$ and ω_{G_i}.
  end for
end for
Evaluate FID and $\frac{1}{IS}$.
$\mathcal{P}'^t_D \leftarrow$ Sort and select the non-dominating individuals on validation set.
$\mathcal{P}_D^t \leftarrow$ Crossover and mutation operator.
$\omega_{G_1} = \omega_{G_2} = \cdots = \omega_{G_P} = \omega_{G^*} \leftarrow$ Weight-resets for the generator weights of G^*.
$D^* \sim (m^*, \omega_{D^*}) \leftarrow$ Obtain the optimal discriminator D^* with architecture m^* and weights ω_{D^*}.
return The best GAN architecture (G^*, D^*).

During the first G search phase, we search G architectures with a fixed discriminator $\bar{D}$ by adopting an $N(G)$-to-1(D)

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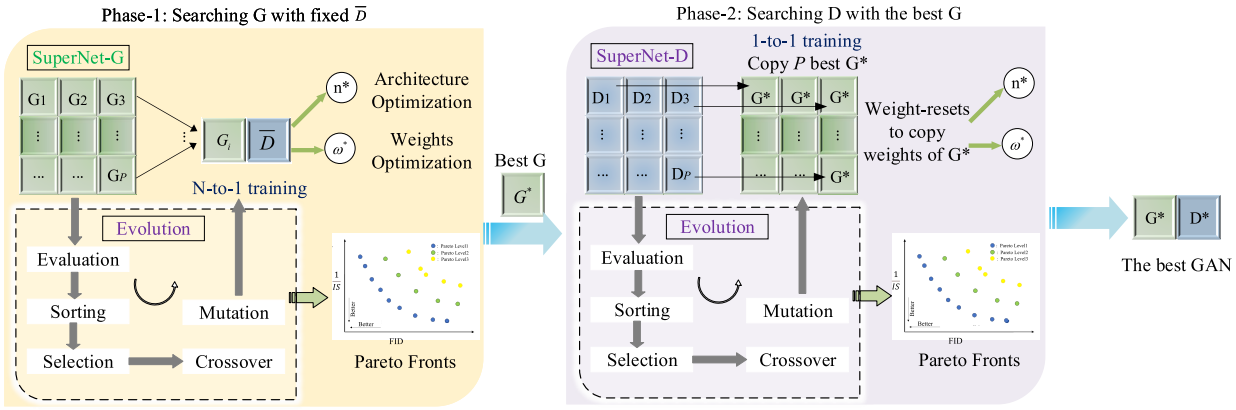

Fig. 3. EGAN architecture search in double phases.

strategy, as shown in Fig. 3, where the N-to-1 denotes all the candidate generators share the same $\bar{D}$. Each generation constructs a population for G architecture evolution $\mathcal{P}_G$, which contains P candidate genotypes of G architectures, where all candidate genotypes share the weights W_G, while each candidate G_i depends on architecture parameter n_i and weights parameter ω_{G_i}. Combine each G_i with the fixed $\bar{D}$ to construct various GAN architectures, so that the corresponding parameters can be described as follows:

$$\begin{aligned}
\omega^*_{G_i} &= \arg\min_{\omega_{G_i}} E_{z\sim p(z)}\left[\log\left(1-\bar{D}(G_i(z))\right)\right], \quad i \in \{1,\ldots,P\} \\
\omega^*_{\bar{D}} &= E_{z\sim p(z)}\left[\log\left(1-\bar{D}(G_i(z))\right)\right] \\
&\quad + \arg\max_{\omega_{\bar{D}}} \sum_{i=1}^{P} E_{x\sim p_{\text{data}}(x)}\left[\log\left(1-\bar{D}(G_i(z))\right)\right] \\
n^* &= \arg\min_{n_i} S_{\text{val}}\left(n_i \middle| \omega^*_{G_i}, \omega^*_{\bar{D}}, \bar{m}\right)
\end{aligned} \tag{11}$$

where $\omega^*_{G_i}$ and $\omega^*_{\bar{D}}$ are the weights of the ith G and fixed D, while n^* is the optimal variable of G architecture in terms of the valued function on validation set.

During the second D search phase, the discovered generator G^* with architecture variable n^* acquired in the first phase is employed to guiding the D architecture search. Different with generators, discriminators lack of direct evaluation metrics, as the accuracy of discriminators is unable to be reflected by the whole performance of GANs [53]. Thus, the 1-to-1 training strategy is applied for the D architecture search, which denotes that multiple copies of the G^* is created, and each encoded discriminator in SuperNet-D is paired with a copied $G^{*\prime}$ and then trained independently. For each evolutionary generation, the population of SuperNet-D is $\mathcal{P}_D$, so that P GANs composed of P candidate D and P copied $G^{*\prime}$ can be acquired, where ${G_i}^* \sim (n^*, \omega^*_{G_i})$ and $D_i \sim (m_i, \omega_{D_i})$. The architecture and weights variables can be calculated as follows:

$$\begin{aligned}
m^* &= \arg\min_{m_i} S_{\text{val}}\left(m_i \middle| \omega^*_{G_i}, \omega^*_{D_i}, n^*\right), \quad i \in \{1,\ldots,P\} \\
\omega^*_{G_i}, \omega^*_{D_i} &= \min_{\omega_{G_i}} \max_{\omega_{D_i}} E_{x\sim p_{\text{data}}(x)}[\log D_i(x)] \\
&\quad + E_{z\sim p(z)}\left[\log(1-D_i(G_i(z)))\right].
\end{aligned} \tag{12}$$

Since P generators are trained in an independent way, every generator has the different weights after a generation of 1-to-1 training. To avoid the biased estimation caused by mode collapse, a weight-resets strategy is introduced to first duplicate the weights of the best G $\omega'_{G_i{}^*}$ in the current generation, and then use $\omega'_{G_i{}^*}$ to initialize all Gs in the next generation. The weight-resets tactic makes a fair comparison between candidate Ds and reduces the instability of GAN training.

After weights training, both of the two phases adopted the same evolution algorithm to search both G and D architectures. For the two fitness functions, IS is inversely correlated with FID, so that we calculate FID and (1/IS) for evolution. With respect to the multiobjective optimization, a fast non-dominated sorting (NSGA-II) [54] strategy is applied to achieve the Pareto-optimal individuals. The total process of the evolutionary architecture search in double phases has been shown in Algorithm 2. Each evolutionary round consists of evaluation, sorting and selection, and crossover and mutation operations.

a) Evaluation: The population of each round contains P individuals indicating the genotypes of GAN architectures. Train the candidate GANs to evaluate each of the individuals in the current population $\mathcal{P}_t$ of the tth round, and return the values of two fitness functions. To reduce the evaluation time, a low-fidelity evaluation strategy is adopted to decrease the number of images for calculating FID and IS.

b) Sorting and Selection: Combine the parent populations and offspring populations to obtain a union population $\mathcal{U}_t$. In terms of the fitness values, rank the individuals in population $\mathcal{U}_t$ through the fast non-dominated sorting algorithm to achieve the ordered non-dominated fronts $\mathcal{F}_t = \{F_1, F_2, \ldots, F_k\}$, where k is the front number and F_1 is the best front. Then, introduce the crowding distance strategy to compare the individuals with the same rank. Select the P individuals in order.

c) Crossover and Mutation: The crossover and mutation operator is also executed on one-hot sequence. Set the probability of performing crossover and mutation. For crossover, select two individuals, and exchange a single one-hot sequence. While for mutation, randomly select the individuals, and change the code of specific position in terms of mutation probability.

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3) Fully Train the Discovered Network GASNet: After the evolutionary architecture search in double phases, the GAN architecture GASNet of (G^*, D^*) with the best performance can be discovered at last. Then, input the data with positive changed, negative changed, and unchanged labels to fully train the GASNet for generating sample figures by category. Specifically, additional label information y is embedded into discriminator G^* and generator D^* as an additional input layer. Thus, the loss function of GASNet can be rewritten as follows:

$$\min_G \max_D V(G, D) = \mathbb{E}_{x\sim p_{\text{data}}}\left[\log D(x|y)\right] + \mathbb{E}_{z\sim p_z}\left[\log(1 - D(G(z|y)))\right]. \quad (13)$$

By fully training the discovered GASNet, reliable training labeled samples of positive changed class, negative changed class, and unchanged class can be produced for the subsequent M-nary CD task. For labeled sample generation, we take the discovered GASNet as a pretraining model and utilize all of the available datasets to manufacture labeled samples for training GASNet, so that labeled samples of each type can be produced by GASNet in advance, and then, we can directly use the supplementary samples to execute the CD network for each specific dataset. Besides, the pretraining model GASNet training with all the available datasets also lifted restrictions using only a few images in one dataset.

C. Convolution-Based BCD and TCD

As mentioned above, a robust DI has been obtained via SNR operator. To automatically determine the number of changed classes, an histogram-based threshold strategy has been proven to be effective in [8] by determining automatically the number of decision thresholds with cost function analysis. Specifically, the histogram of DI can be defined as follows:

$$h(k) = P(v_u)p(k|v_u) + \sum_{i=1}^{2} P(v_{c_i})p(k|v_{c_i}) \quad (14)$$

where v_u, v_{c_1}, and v_{c_2} represent the distributions of unchanged changed, negative changed, and positive changed classes, respectively. $P(v)$ is the prior probability, while $p(k|v)$ is the class-conditional density. Then, the double-threshold KI criterion function can be formulated as follows:

$$J(T_1, T_2) = \sum_{k=1}^{L-1} h(k)c(k, T_1, T_2) \quad (15)$$

$$c(k, T_1, T_2) = \begin{cases} -2\ln P^{c_1}(k, T_1, T_2), & \text{if } k \leqslant T_1 \\ -2\ln P^{u}(k, T_1, T_2), & \text{if } T_1 \leqslant k \leqslant T_2 \\ -2\ln P^{c_2}(k, T_1, T_2), & \text{else.} \end{cases} \quad (16)$$

Then, minimize the $J(T_1, T_2)$ to obtain the optimal threshold values. Finally, the number of changed categories can be calculated by the Hessian matrix

$$H_{(T_1^*, T_2^*)} = \begin{bmatrix} \dfrac{\partial^2 J}{\partial T_1^2} & \dfrac{\partial^2 J}{\partial T_1 \partial T_2} \\ \dfrac{\partial^2 J}{\partial T_1 \partial T_2} & \dfrac{\partial^2 J}{\partial T_2^2} \end{bmatrix}, \quad T_1^* < T_2^* \quad (17)$$

where if $H_{(T_1^*, T_2^*)}$ is positive definite, then both negative and positive changed classes exist in the SAR image pair; else if $(\partial^2 J/(\partial T_1^2)) \leqslant 0$, then only negative changed class exists; if $(\partial^2 J/(\partial T_2^2)) \leqslant 0$, then only positive changes exist.

On the other side, in order to make the input data of the classifier, we segment the DI into superpixels. Compared with grid cropped patches, a superpixel is a group of pixels with adjacent space and similar backscattering coefficient, which contains more plentiful information and spatial relationships. Therefore, a simple linear iterative clustering (SLIC) [55] algorithm is applied to obtain uniform and compact superpixels. Then, a mirror-filling operator is applied to make superpixels in fixed and regular shape. According to the number of classes, the CD task can be further divided into BCD and TCD. Then, with respect to the specific task, the GASNet generates the corresponding category labels to expand the training sample set. Each classifier contains two convolutional modules and two FC layers, where the convolutional module consists of a convolutional, a batch norm layer, and a rectified linear unit (ReLU) layer. The convolutional layers process 2-D spatial contextual information, while the bottom FC layer outputs the category label of each superpixel patch. The overall loss function of classifier can be defined as follows:

$$L = \frac{1}{n}\sum_{k=1}^{n}\sum_{i=1}^{C} L\left(y_i^k, f_i^k\right) \quad (18)$$

where n denotes the numbers of mini-batch in labeled data for Adam optimizer and f_i^k represents the output unit of the k sample in labeled data with label y_i^k. Besides, $L(y_i^k, f_i^k)$ is in the form of cross entropy.

IV. Experimental Settings

In this section, the descriptions to all SAR datasets are ready for use, and the evaluation criteria will be elaborated. Then, an introduction to the involved comparison algorithms and parameter settings will be presented.

A. Datasets

To evaluate the effectiveness of the proposed GASCD on both BCD task and TCD task, we select four datasets containing positive and negative changes for verifying the performance on TCD task and three datasets with single changed types for BCD task. The datasets on TCD task contain a set of synthetic dataset and three real SAR image datasets. The synthetic dataset consists of seven pairs of synthetic SAR images with different equal number of looks (ENL), where ENL represents the degree of noise erosion (a lower ENL value denotes a higher noise level), which is shown in Fig. 4. Analysis on the synthetic images with different ENLs could explore the noise robustness of methods. Besides, the first real SAR image pair is obtained by Radarsat-2 over the Yellow River of China, which reflects the changes about the river's estuary area. With the size of 291 × 444 and the spatial resolution of 8 m, the temporal images are captured in June 2008 and June 2009, respectively, as shown in Fig. 5(a) and (b). Fig. 5(c) is the ground-truth image formed via artificial marking by expert knowledge and on-the-spot

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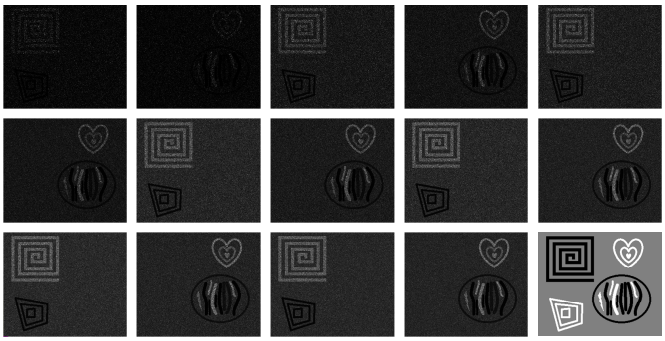
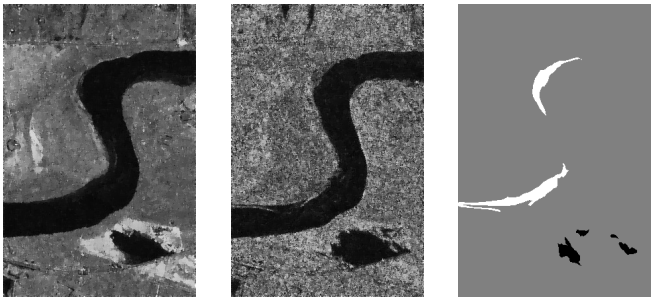
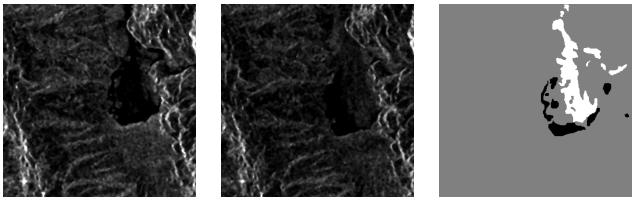


Fig. 4. Synthetic dataset with seven different ENLs (ENL = 1–7).



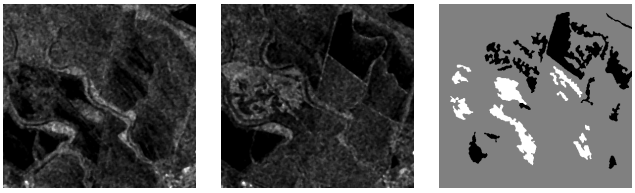
(a) (b) (c)

Fig. 5. YellowRiver dataset and the reference map for TCD. (a) Image acquired on June 2008. (b) Image acquired on June 2009. (c) Ground truth.



(a) (b) (c)

Fig. 6. San Francisco-1 dataset and the reference map. (a) Image captured in August 2003. (b) Image captured in May 2004. (c) Ground-truth image.

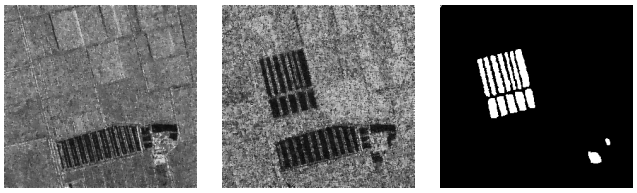


(a) (b) (c)

Fig. 7. San Francisco-2 dataset and the reference map. (a) Image captured in August 2003. (b) Image captured in May 2004. (c) Ground-truth image.

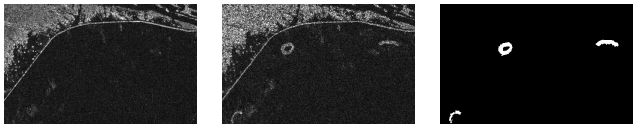
investigation. In Fig. 5(c), the white regions denote positive changed areas, the black regions are the negative changed areas, and the gray regions are the unchanged areas. The second and third real datasets used for TCD task are two SAR image pairs from the European Remote Sensing Satellite-2 of European Space Agency (ESAERS-2) satellite over the city of San Francisco, in August 2003 and May 2004, respectively. The size of the second dataset shown in Fig. 6(a) and (b) is 390 × 390 with the spatial resolution of 26 m, while the third real dataset shown in Fig. 7(a) and (b) has the size of 390 × 370. Both of the ground-truth images are generated through surface prior information.

In addition, we also apply three real SAR datasets for verifying algorithm performance on BCD task. The first and



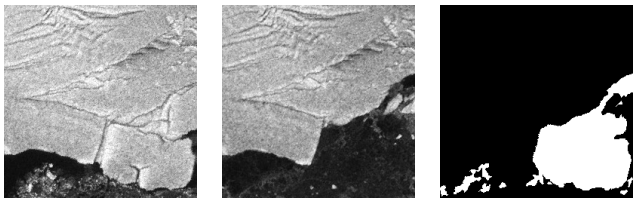
(a) (b) (c)

Fig. 8. Farmland dataset and the reference map for BCD task. (a) Image captured in June 2008. (b) Image captured in June 2009. (c) Ground-truth image.



(a) (b) (c)

Fig. 9. Coastline dataset and the reference map for BCD task. (a) Image captured in June 2008. (b) Image captured in June 2009. (c) Ground-truth image.



(a) (b) (c)

Fig. 10. Sulzberger dataset and the reference map for BCD task. (a) Image captured on March 11, 2011. (b) Image captured on March 16, 2011. (c) Ground-truth image.

second BCD datasets are also taken by Radarsat-2 satellite at the region of Yellow River estuary. With the size of 306 × 291, the Farmland dataset shown in Fig. 8 records the changes of farmland in the Yellow River basin. The second dataset named Coastline dataset is of 450 × 280 pixels reflecting the changes of coastal intertidal zone near the Yellow River estuary, which can be shown in Fig. 9. The third SAR dataset for BCD is Sulzberger dataset, which was taken by the European Space Agency's ENVISAT satellite on March 11 and 16, 2011 at the region of the Sulzberger ice shelf. As shown in Fig. 10, the Sulzberger dataset has a section size of 256 × 256 showing the zone of the ice breakup.

B. Evaluation Criteria

For the mainstream binary CM, there are five popular metrics applied to quantitatively evaluate the performance, i.e., false negative (FN), false positive (FP), overall errors (OEs), percentage correct classification (PCC), and Kappa coefficient (KC). To evaluate the ternary CM, Su et al. [38] and Liang et al. [41] improved the metrics derived from BCD task. Here, we apply the following metrics to access the performance of TCD methods.

1) A confusion matrix $\mathbf{P}$ is applied to count pixels of all categories of whether classified correctly or not

$$\mathbf{P} = \begin{bmatrix} T_{uu} & F_{up} & F_{un} \\ F_{pu} & T_{pp} & F_{pn} \\ F_{nu} & F_{np} & T_{nn} \end{bmatrix}. \tag{19}$$

Then, the following 2)–4) are introduced to simplify $\mathbf{P}$.

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2) Positive missed alarms (PMAs)

$$\text{PMA} = F_{\text{pu}} + F_{\text{pn}}. \tag{20}$$

3) Negative missed alarms (NMAs)

$$\text{NMA} = F_{\text{nu}} + F_{\text{np}}. \tag{21}$$

4) Positive false alarms (PFAs)

$$\text{PFA} = F_{\text{up}} + F_{\text{np}}. \tag{22}$$

5) Negative false alarms (NFAs)

$$\text{NFA} = F_{\text{un}} + F_{\text{pn}}. \tag{23}$$

6) Besides, the OE for TCD

$$\text{OE} = N - T_{\text{uu}} - T_{\text{pp}} - T_{\text{nn}} = N - \text{tr}(\mathbf{P}) \tag{24}$$

where N is the total number of pixels.

7) PCC for TCD

$$\text{PCC} = \frac{T_{\text{uu}} + T_{\text{pp}} + T_{\text{nn}}}{N} = \frac{\text{tr}(\mathbf{P})}{N}. \tag{25}$$

8) KC to evaluate the classification accuracy on TCD

$$\text{KC} = \frac{N \cdot \text{tr}(\mathbf{P}) - \mathbf{t}\mathbf{P}^2\mathbf{t}^{\mathbf{T}}}{N - \mathbf{t}\mathbf{P}^2\mathbf{t}^{\mathbf{T}}} \tag{26}$$

where $\mathbf{t} = [1, 1, 1]$. KC is a statistical measure of the accuracy or consistency of the difference between the error matrix and the chance consistency. A larger KC indicates a better performance of classification.

C. Comparison Algorithms and Parameter Settings

In terms of comparison algorithms, we set up multiple popular methods for TCD and BCD tasks, respectively. With respect to ternary SAR image CD task, we choose a threshold algorithm [GG-GKI thresholding (GKIT)] [14] as well as a clustering algorithm [17] (FLICM), and then combined them with the proposed DI generation method (SNR) as two comparison methods. Then, an evolutionary-based unsupervised method multiobjective Gaussian mixture model (MOGMM) [56] is introduced for TCD task, which utilizes the multiobjective optimization strategy to optimize the Gaussian mixture model. The SAE + CNN method proposed in [39] combines the SAE and CNN to solve ternary changed detection problem. Since the advanced works available on TCD are relatively few, we also have improved three effective BCD algorithms [NR and extreme learning machine (NR-ELM) [57], PCANet [34], and Ms-CapsNet [58]] to execute the ternary changed detection task. NR-ELM applies the ELM model to classify the pixels of DI. PCANet utilizes PCA filters as convolutional filters when it combines PCANet model results and the preclassification results to form the final CM. Besides, Ms-CapsNet is an advanced supervised SAR image CD method, which uses a multiscale capsule network with an adaptive fusion convolution to extract the discriminative information while considering the spatial relationship of features. For more convenient comparison, we also applied the above seven algorithms as comparison on binary SAR image CD task. Besides, we also add a frequency domain-based CD approach convolutional-wavelet neural network (CWNN) [2], where the CNN classifier uses dual-tree complex wavelet transform to reduce the effect of speckle noise.

Except the DI generation step by SNR operator is executed on MATLAB R2020, all the other steps are implemented by Python 3.8 on double GeForce RTX 3090 GPUs. In the SNR operator, we set the adjust coefficient α to be 3. For the neighborhood size S of SNR operator, we set the size of S as 3×3, 5×5, and 9×9, and generate their DIs, respectively. By comparing their AUC values, we can observe that the size of S as 5×5 made the best performance.

For the GASNet in the label sample generation step, the input data are the image patches acquired by all the available SAR image-pair datasets. The GASNet search stage utilized the Adam optimizer and the hinge loss with a momentum of 0.9 and a learning rate of 0.002. Besides, the population size P is set to 32, and the batch sizes are 40 and 80 for generator and discriminator, respectively. All the P candidate individuals are uniformly activated and equally trained. The learning rate, momentum, and batch size of full trained stage are the same as the search stage, but the number of training epochs is set to 500, while the momentum is set as 0.9.

For the BCD and TCD network, each convolution-based classifier consists of two convolution layers and two bottom FC layers, where each convolution layer possesses the 3×3 kernel size with a batch normal layer and a ReLU layer, which is also optimized by the Adam optimizer. Then, we carry out two parameter tests for searching the suitable kernel number K_n and learning rate r.

V. Experimental Results

A. Analysis of the SNR Operator for DI Generation

In order to demonstrate the influence of the proposed SNR operator in the DI generation process, we compare the SNR operator with two popular DI generation operators for TCD, LR operator and MR operator. We apply the three operators on the synthetic dataset and three real SAR TCD datasets to generate the DIs and draw their respective receiver operating characteristic (ROC) curves. In terms of the ROC curves, the area under curve (AUC) values can be calculated to reflect the performance of generated DIs intuitively. Fig. 11 shows the generated DIs of three operators on the synthetic dataset and their ROC figures. From Fig. 11(a)–(g), the ENL gradually increases, which denotes that the erosion degree of noise gradually decreases.

As shown in Fig. 11, DIs produced by LR exist a large amount of noise, while DIs produced by MR omit part of change information, such as the top-right portion of each DI. Compared with LR and MR, DIs produced by SNR could better describe the difference information. From the ROC curves, we could also find the SNR curve is always on the top of LR and MR curves, which demonstrates the superiority of the proposed SNR operator for DI generation. Besides, from Fig. 11(d)–(g), we can find that ROC curves of MR are close to those of SNR curves when the noise is less. But, according to Fig. 11(a)–(c), we find that as the noise erosion increases, the performances of LR and MR are significantly degraded, while SNR still maintains in operating condition, which denotes that

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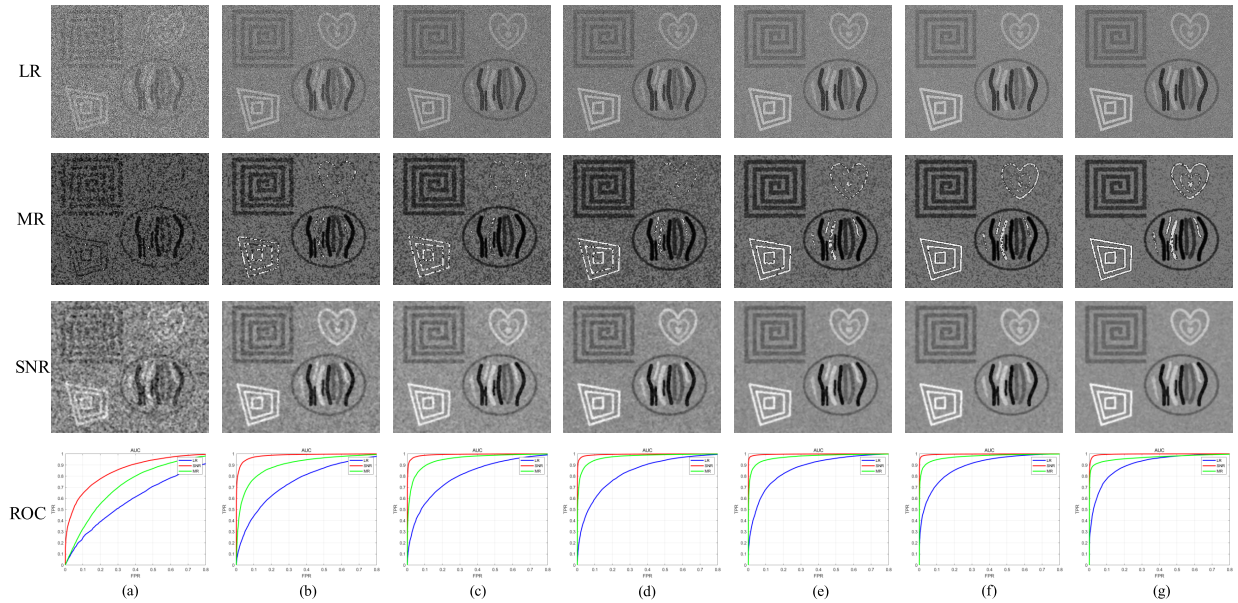

Fig. 11. Different images generated by three operators and their ROC curves on the synthetic dataset with different ENLs. (a)–(g) DIs on the synthetic image pair with ENL = 1–7 as well as their ROC curves, respectively.

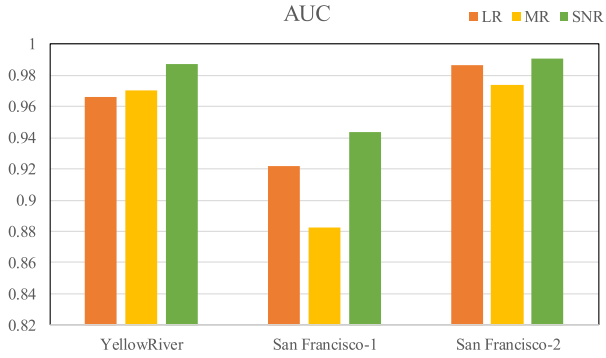

Fig. 12. AUC values of DIs generated by LR, MR, and SNR operators on three real SAR datasets.

TABLE I

RELATIONSHIP BETWEEN PCC (%) VALUES AND NUMBERS OF SUPPLEMENTED SAMPLES GENERATED BY GASNET

Dataset	0%	5%	10%	15%	20%	25%
YellowRiver	98.01	98.24	98.4	98.75	98.86	98.84
San Francisco-1	93.01	94.98	95.66	96.23	96.63	96.98
San Francisco-2	95.46	96.90	97.12	97.58	97.92	97.96

the SNR operator has strong robustness to noise. In addition, we also experiment three operators on three real SAR datasets and calculate their AUC values of ROC curves, which are shown in Fig. 12. DIs produced by the SNR operator has the highest AUC values on all the SAR datasets, which also adds to the proof that the proposed SNR operator has made a good performance on the DI generation task.

B. Analysis of the GAN Architecture Search-Based Sample Generation Strategy

1) Influence of the Supplemented Samples: In order to verify whether the supplemented samples generated by GASNet are directly effective for the final CD task, we set a comparative experiment on the number of supplemented samples N. For the comparative experiment, we set different numbers of the supplemented samples generated by GASNet on three real SAR datasets and record their corresponding PCC and KC values of CMs. We set a discrete search interval of $N = [0\%, 5\%, 10\%, 15\%, 20\%, 25\%]$, where N denotes the ratio of supplementary samples number to the total patch amount of the SAR image pair to be tested. In the interval, $N = 0$ means only the limited original labeled samples are input without using the GASNet to produce any samples. During the supplemented samples, positive changed labels, negative changed labels, and unchanged labels each account for one-third of the supplemented data. Table I reports the relationship between PCC (%) values and numbers of supplemented samples, while Table II reports the relationship between KC values and supplemented sample numbers. PCC is used to measure the overall accuracy of CM. From Table I, we find that as the number of supplemented samples increases, the accuracy of CD results on all the three datasets supplemented gradually. Especially on the two San Francisco datasets, the upward trend is particularly obvious. For the YellowRiver dataset, the upward trend is relatively gentle, but as shown in Table II, we can observe that the KC values of three datasets are improved significantly when the number of samples increases. As a measure of consistency, KC can better measure the classification effects of different categories when the sample categories are unbalanced. For the TCD task, the numbers of positive changes and negative changes are far less than the unchanges. The significant improvement of

TABLE II

RELATIONSHIP BETWEEN KC VALUES AND NUMBERS OF SUPPLEMENTED SAMPLES GENERATED BY GASNET

Dataset	0%	5%	10%	15%	20%	25%
YellowRiver	0.6967	0.7465	0.7638	0.7981	0.8165	0.8102
San Francisco-1	0.7109	0.7805	0.8223	0.8518	0.8602	0.8689
San Francisco-2	0.6472	0.7433	0.7887	0.8214	0.8320	0.8377

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TABLE III
QUANTITATIVE COMPARISON ON DIFFERENT GAN ARCHITECTURES

Models	IS ↑	FID ↓	YellowRiver Dataset		San Fran-1 Dataset		San Fran-2 Dataset	
			PCC(%)	KC	PCC(%)	KC	PCC(%)	KC
DCGANS [59]	0.89±0.18	13.7	4427	0.6127	9402	0.6046	8836	0.7801
WGAN [30]	0.98±0.13	11.4	2561	0.7218	5493	0.7126	7990	0.7972
BigGAN [51]	1.23	9.8	1607	0.7399	4857	0.7455	7217	0.8140
AGAN [45]	1.01±0.07	10.8	2320	0.7017	5503	0.7184	7720	0.8089
AutoGAN [44]	1.08±0.10	9.88	2146	0.7575	5398	0.7265	6960	0.8192
E2GAN [46]	1.18±0.11	9.13	1934	0.7716	3679	0.8003	6200	0.8431
AdversarialNAS [48]	1.08±0.08	8.91	1512	0.8140	3502	0.8093	6681	0.8290
GASNet	1.24±0.10	8.55	1489	0.8143	3172	0.8308	5212	0.8676

TABLE IV
ABLATION STUDY OF EACH MODULE OF GASCD

Methods	PMA	NMA	PFA	NFA	OE	PCC(%)	KC
Module A	2013	722	1386	3348	7469	95.09	0.6610
Module A+B	2231	648	955	468	4302	97.17	0.7740
GASCD(A+B+C)	1632	751	632	157	**3172**	**97.91**	**0.8308**

KC values shown in Table II demonstrates that the increase of the supplemented samples leads to a better classification performance of positive changed class, negative changed class, as well as unchanged class. Both PCC and KC values indicate that the proposed GAN-based sample generation strategy not only achieves a higher accuracy of CD but also a more excellent classification effects on each changed type.

2) Analysis of the GAN Architecture Search Technique: In an effort to further explore the effectiveness of the proposed GAN architecture search-based sample generation strategy, we compare the proposed GASNet with some GANs and several advanced NAS-GAN techniques. In order to more intuitively evaluate the sample generation ability of different algorithms, we select IS and FID score as two indicators to reflect the reliability and diversity of generated sample patches. Since most of NAS-GANs are devoted to unconditional image task, we abandoned the label information of the proposed GASNet so as to compete fairly with other algorithms. We experimented these models on our real SAR CD datasets and recorded the IS, FID scores, and GPU days in Table III. A higher IS means a greater reliability of generated images, while a lower FID means a better diversity. From Table III, it can be observed that our proposed GASNet achieved the highest IS value and least FID score, which represents that our GASNet possesses a better ability of generating reliable samples. To explore the sensitivity of GANS, we trained GANS 20 times and recorded the metrics for each training session. The experimental results show that the IS values of the label samples generated by 20 training sessions are within the range of [1.14, 1.34]. Then, the PCCs of CMs obtained 20 times on three SAR dataset are stable between 98.85% ± 0.08%, 97.91% ± 0.10%, and 96.39% ± 0.12%, which demonstrates the stability of the labeled samples generated by the discovered GANS model.

TABLE V
QUANTITATIVE COMPARISON AMONG EACH MODEL ON THE YELLOWRIVER DATASET

Methods	PMA	NMA	PFA	NFA	OE	PCC(%)	KC
SNR+GKIT [14]	636	186	1213	565	2600	97.99	0.7171
SNR+FLICM [17]	255	0	415	41387	42057	67.45	0.1258
PCANet [34]	285	3	1917	37568	39772	69.22	0.1322
NR-ELM [57]	320	703	3272	5	4299	96.67	0.5865
SAE+CNN [39]	719	820	759	0	2298	98.22	0.6950
MOGMM [56]	1003	104	258	562	1927	98.51	0.7597
MsCapsNet [58]	1185	302	132	222	1841	98.58	0.7448
GASCD	674	210	517	88	**1489**	**98.85**	**0.8143**

TABLE VI
QUANTITATIVE COMPARISON AMONG EACH MODEL ON THE SAN FRANCISCO-1 DATASET

Methods	PMA	NMA	PFA	NFA	OE	PCC(%)	KC
SNR+GKIT [14]	2535	783	1431	5812	10498	93.10	0.5583
SNR+FLICM [17]	2013	722	1405	3361	7469	95.09	0.6610
PCANet [34]	1968	631	1809	8234	12540	91.76	0.5345
NR-ELM [57]	2328	1846	1495	499	6157	95.95	0.6620
SAE+CNN [39]	2876	917	650	2606	7038	95.37	0.6436
MOGMM [56]	1868	790	1513	3051	7203	95.26	0.6713
MsCapsNet [58]	1392	1605	1576	350	4894	96.78	0.7450
GASCD	1632	751	632	157	**3172**	**97.91**	**0.8308**

Besides, in order to further access the contribution of the GAN architectures to CD, we also reported the accuracies of CD by employing different GAN architectures. Compared with other GAN architectures, our proposed GASNet achieves the best OE and KC values on all the three datasets, which indicated that the proposed NAS-based GAN architecture is better equipped for the M-nary SAR image CD tasks. Moreover, to further explore the computational complexity of our GASNet, we calculate the Params of it with state-of-the-art (SOTA) models. The relationship between Params and IS as well as FID is shown in Fig. 13, where we can observe that our GASNet could achieve the best IS and FID values with a smaller Params than other GAN models.

3) Ablation Study of the Decoupled Search Process: For the sake of verifying our decoupled search strategy, we conducted a coupled search experiment as a baseline; i.e., it evolves the architectures of generator and discriminator concurrently in each search generation. Then, we recorded the changes of IS and FID scores in Fig. 14, which reflects the learning curves of the coupled baseline and our decoupled GASNet. In our decoupled search strategy, the first 100 epochs are the generator search phase, while the next 100 epochs belong to

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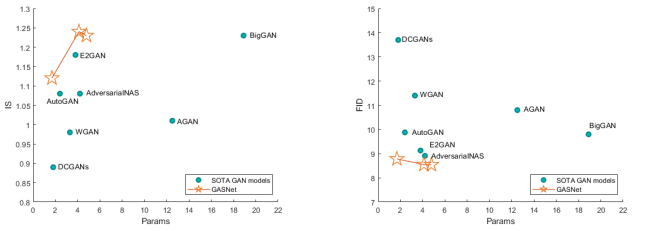
TABLE VII
QUANTITATIVE COMPARISON AMONG EACH MODEL ON THE SAN FRANCISCO-2 DATASET

Methods	PMA	NMA	PFA	NFA	OE	PCC(%)	KC
SNR+GKIT [14]	2237	2582	910	3145	8869	93.85	0.7789
SNR+FLICM [17]	188	3030	5243	1486	9947	93.11	0.7717
PCANet [34]	180	2326	8977	3240	14720	89.80	0.6953
NR-ELM [57]	276	7227	10215	1234	18949	86.87	0.5707
SAE+CNN [39]	2008	6944	322	322	10097	93.00	0.7112
MOGMM [56]	1437	4626	1042	964	8069	94.41	0.7863
MsCapsNet [58]	1285	2736	576	1742	6339	95.61	0.8394
GASCD	1484	2021	410	1302	**5212**	**96.39**	**0.8676**



(a) (b)

Fig. 13. (a) Relationship between Params and IS of different GAN models. (b) Relationship between Params and FID of different GAN models.

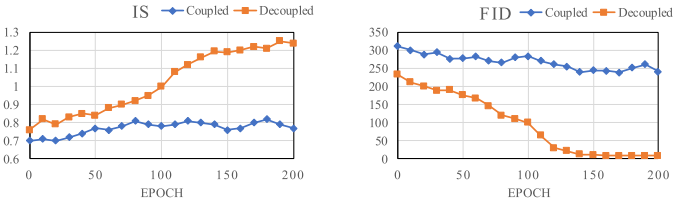


Fig. 14. Learning curves of our decoupled search process and the coupled search baseline.

the discriminator search phase. From Fig. 14, we can observe that the coupled search is unstable throughout the search process. On the contrary, the learning curve of the proposed decoupled search has a clear upward trend. Especially, in the second half of the curve, IS and FID scores of our decoupled search are significantly improved, which indicates that the decoupled search process effectively enhances the stability of GAN training. Besides, our decoupled search always achieved better IS and FID scores than the coupled baseline, demonstrating the decoupled search strategy made a better performance on generating reliable samples.

C. Analysis of Convolutional CD Network

1) Influence of Sample Patch Size SP: In our proposed framework, the input image patches are acquired by superpixels with filling operation, which requires that the original patches and supplemented sample patches generated by GASNet have the same size. The superpixel size directly affects the sample patch size and, thereupon, then has a great impact on the final CD results. We set the sample patch size SP to be 3×3, 5×5, 7×7, 9×9, and 11×11, and obtain the corresponding CD results. The corresponding values of PCC and KC of CD results generated by different superpixel sizes are shown in Fig. 15. It can be seen that as the sample patch increases, the accuracies of CD results

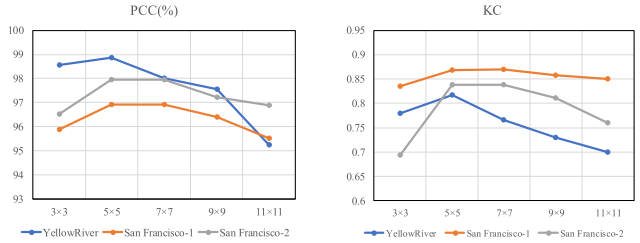
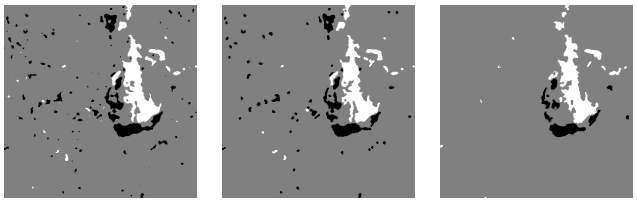


Fig. 15. PCC and KC values on three datasets with different sample patch sizes.



(a) (b) (c)

Fig. 16. CMs of obtained by different combinations of GASNet modules. (a) CM of using Module A. (b) CM of using Modules A and B. (c) CM of the proposed GASCD (Modules A–C).

first increase but then decrease. When the parameter SP is set to be smaller, superpixels become smaller, and some important context information may be lost. But, a larger SP will blur the objects' boundaries, lose the detail information, and reduce the diversity of training samples. As shown in Fig. 15, the YellowRiver dataset achieved the best results when $SP = 5 \times 5$, but then the accuracy dropped sharply as SP decreases. For the other two San Francisco datasets, both $SP = 5 \times 5$ and 7×7 could make a good CD performance. Considering about noise abatement and abundant context information, we uniformly select $SP = 5 \times 5$ as the final sample patch size.

2) Parameter Analysis: Both the convolutional-based BCD and TCD networks have the same network architecture, which contain two convolutional modules and two FC layers, where the convolutional kernels play an important role in the abstract feature extraction [60]. Thus, we conducted a comparison experiment with different numbers of convolutional kernels. Experiment results show that when the number of the convolutional kernels in the first layer is 12 and that in the second layer is 24, the results achieve the best performance on PCC and Kappa. In addition, the learning rate r is also a crucial parameter of network training performance. We vary r on a logarithmic scale [0.1, 0.01, 0.001] and then report their results. The testing results find that a smaller learning rate could make a better CD performance. Thus, we set the value of learning rate to 0.001 for the proposed model in follow-up experiments.

3) Ablation Study of Each Module: For verifying each module of the proposed method, we employed (A) difference information representation module with clustering algorithm, using the training CD classifier (C) trained with the pseudo-labels obtained by (A) module, and finally employing the (B) NAS-based labeled sample generation GASNet module with (A) and (C), respectively, to produce the CMs. Fig. 16

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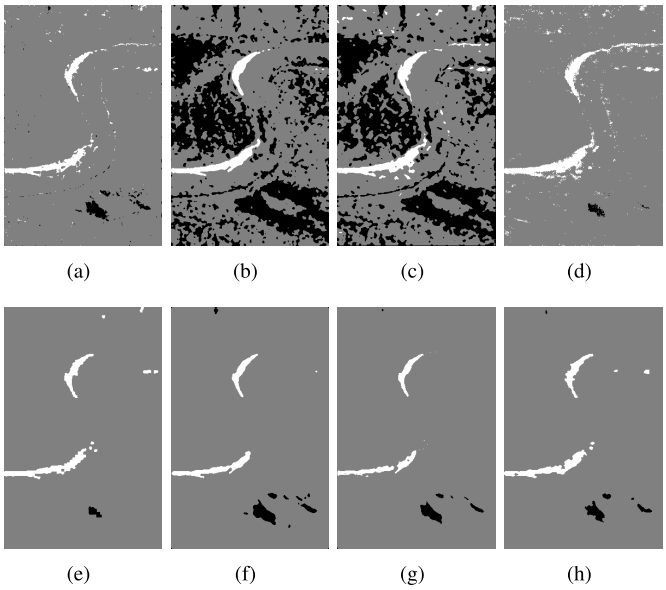

Fig. 17. Ternary CMs of different methods on the YellowRiver dataset. (a) SNR + GKIT. (b) SNR + FLICM. (c) PCANet. (d) NR-ELM. (e) SAE + CNN. (f) MOGMM. (g) MsCapsNet. (h) Proposed GASCD.

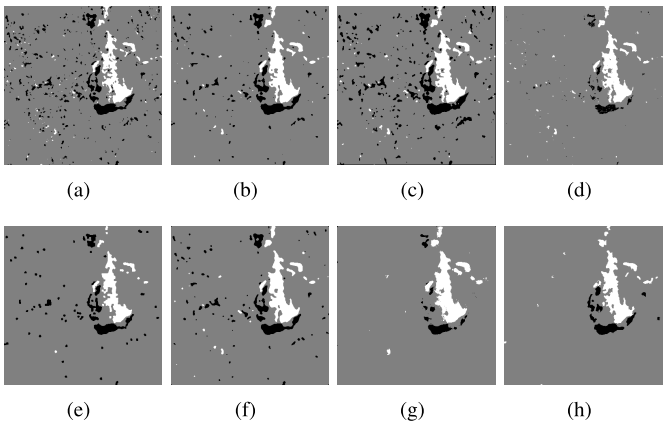

Fig. 18. Ternary CMs of different methods on the San Francisco-1 dataset. (a) SNR + GKIT. (b) SNR + FLICM. (c) PCANet. (d) NR-ELM. (e) SAE + CNN. (f) MOGMM. (g) MsCapsNet. (h) Proposed GASCD.

shows the different CMs obtained by the above three methods, where we can obviously find that the CM obtained by using Modules A–C has the less noise areas, which indicated that the employment of NAS-based labeled sample generation strategy and convolutional CD network are helpful to learn the difference features. Quantitative analysis listed in Table IV also demonstrates that using Modules B and C is able to improve the accuracy of M-nary CD.

D. Detection Performance

To validate the effectiveness of the proposed M-nary CD framework GASCD, we first compare GASCD with some advanced approaches mentioned in Section IV on the three real SAR datasets for TCD task. The visual results and the statistical values of PFA, NFA, NMA, PMA, OE, PCC, and KC are reported on each data. Besides, to validate the ability of GASCD for handling only one changed category, we also experiment GASCD with other eight BCD approaches on three

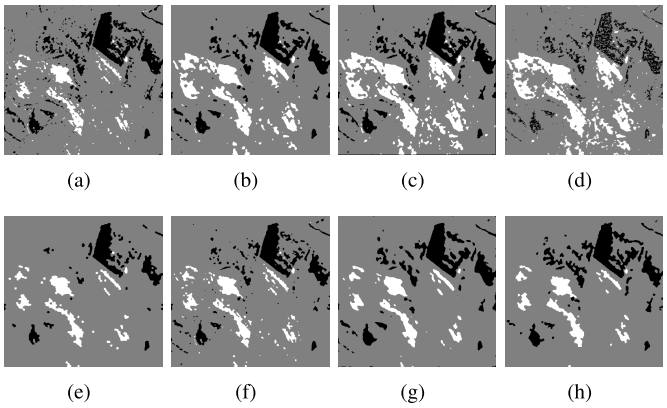

Fig. 19. Ternary CMs of different methods on the San Francisco-2 dataset. (a) SNR + GKIT. (b) SNR + FLICM. (c) PCANet. (d) NR-ELM. (e) SAE + CNN. (f) MOGMM. (g) MsCapsNet. (h) Proposed GASCD.

SAR datasets containing a certain changed type. In the same way, the final CMs and statistical values have been recorded.

1) Experiment on the YellowRiver Dataset: The first TCD experiment is carried out on the YellowRiver dataset, which is composed of positive changed class (white zone), negative changed class (black zone), and unchanged class (gray zone). Both positive and negative changed regions only make up a small portion of the total image. Fig. 17 shows the ternary CMs of the YellowRiver dataset obtained by different methods. As shown in Fig. 17, the CM acquired by SNR + FLICM falsely detected a large area of unchanged as negative changed regions, which indicates that the unsupervised clustering method is unable to differentiate the characteristics between negative changed pixles and unchanged pixels on this dataset. Since the PCANet heavily relies on the pre-classified results and we apply the PCA-Kmeans algorithm for the pre-classification of PCANet, the unsupervised clustering of pre-classification results in a terrible CD result. Besides, except for the SNR-GKIT and NR-ELM algorithms existing many white noise, the remaining four methods are able to detect positive changed regions fairly well. However, due to the lack of enough reliable negative changed samples, SAE + CNN, MOGMM methods cannot correctly detect the negative changed area. By contrast, our proposed GASCD method effectively describes the real zone of negative changes, which benefits from the reliable labeled samples generated by GASNet. The numerical results of evaluation criteria are listed in Table V, where we can find the our GASCD achieved the least OE and the highest PCC and KC. Although the PCCs of SNR-GKIT, MOGMM, and MsCapsNet are close to that of GASCD, the outstanding KC value of GASCD indicates that our GASCD has good classification performance for both positive and negative changed classes.

2) Experiment on the San Francisco-1 Dataset: Fig. 18 shows the experimental results generated by the proposed method and other seven comparative methods on the San Francisco-1 dataset. All of the eight methods can detect the obvious changes (both positive and negative changes). However, the CMs of SNR + GKIT, SNR + FLICM, PCANet, and MOGMM exist lots of black noise, which denotes the unsupervised clustering or threshold is quite

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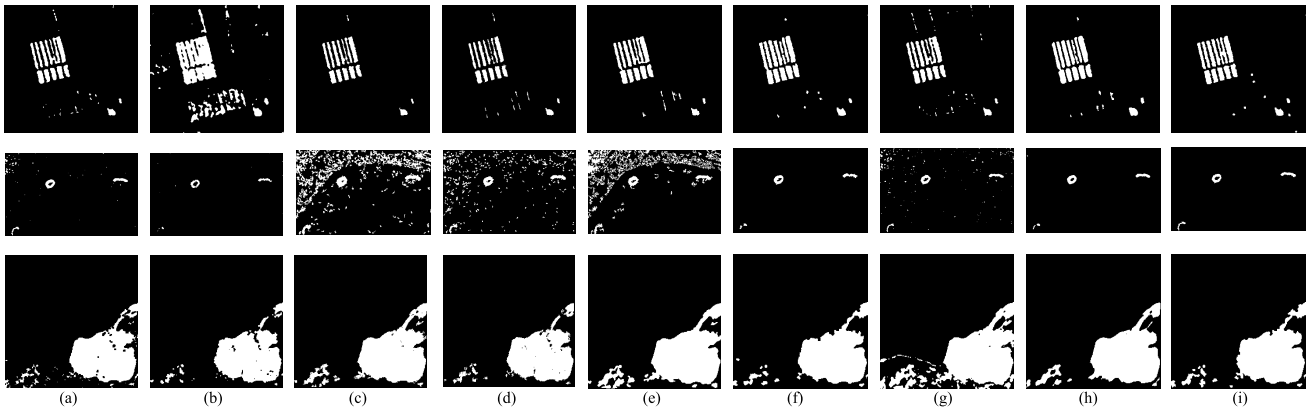

Fig. 20. Binary CMs of different CD methods on three BCD datasets. (a) SNR + GKIT. (b) SNR + FLICM. (c) PCANet. (d) NR-ELM. (e) CWNN. (f) SAE + CNN. (g) MOGMM. (h) MsCapsNet. (i) Proposed GASCD.

TABLE VIII
BINARY CHANGE RESULTS ON THREE SAR DATASETS WITH SINGLE CHANGED TYPE

Methods	Farmland Dataset					Coastline Dataset					Sulzberger Dataset				
	FP	FN	OE	PCC(%)	KC	FP	FN	OE	PCC(%)	KC	FP	FN	OE	PCC(%)	KC
SNR+GKIT [14]	375	1629	2004	97.75	0.7725	240	613	853	99.32	0.6295	187	1579	1766	97.31	0.9095
SNR+FLICM [17]	4388	**99**	4487	94.96	0.6724	25	543	568	99.55	0.7371	**7**	1875	1882	97.13	0.9021
PCANet [34]	**51**	1334	1385	98.44	0.8423	16735	**6**	16741	86.71	0.1207	410	524	934	98.57	0.9540
NR-ELM [57]	182	1676	1858	97.91	0.7839	6258	20	6278	95.02	0.2843	25	1544	1569	97.61	0.9193
CWNN [2]	324	734	1058	98.81	0.8893	13223	51	13274	89.47	0.1468	1085	245	1330	97.97	0.9363
SAE+CNN [39]	210	487	697	99.22	0.9279	24	186	210	99.83	0.9163	50	857	907	98.62	0.9544
MOGMM [56]	429	928	1357	98.48	0.8568	1218	228	1446	98.85	0.6023	2500	**46**	2546	96.12	0.8836
MsCapsNet [58]	602	408	1010	98.87	0.8999	66	243	309	99.75	0.8761	422	400	822	98.75	0.9597
GASCD	210	380	**590**	**99.34**	**0.9396**	**17**	150	**167**	**99.87**	**0.9342**	60	606	**666**	**98.98**	**0.9668**

sensitive to noise. By comparison, the CMs of NR-ELM, MsCapsNet, and GASCD could better suppress the irrelevant noise with the introduction of DNNs. As a supervised algorithm, the MsCapsNet method could detect most of the positive changed and negative changed regions with less noise, but the scarce labels of positive and negative changes still limit the performance of MsCapNet, so that several changed details have not been detected correctly in the CM. Because of the supplement of enough samples of positive and negative samples by GASNet, the proposed GASCD performs better in detail of CM. Quantitatively evaluation values are exhibited in Table VI. The terrific amount of black noise of SNR + GKIT, SNR + FLICM, and PCANet resulted in high OEs. The application of DNN can greatly improve the accuracy of CD result, such as NR-ELM and MsCapsNet. Among these methods, GASCD yields the least OE of 3172 and achieves the highest PCC and Kappa values of 97.91% and 0.8308, which represents the effectiveness of the proposed SNR operator and GAN architecture search-based sample generation strategy.

3) Experiment on the San Francisco-2 Dataset: The third experiment is carried on the San Francisco-2 dataset, where Fig. 19 shows the CMs of each TCD algorithm and Table VII records the evaluation metrics. From Fig. 19, we can observe that the CM obtained by SNR-GKIT exists lots of isolated noise, which is limited by the character of threshold methods. In CMs acquired by SNR-FLICM and PCANet, a lot of positive changed regions were detected by mistake. For the NR-ELM, the main negatived regions could not be detected well, indicating that only using ELM technique is unable to distinguish the difference between changes and unchanges very well in this dataset. Compared with the unsupervised MOGMM, CMs of SAE + CNN and MsCapNet possess less isolated pixels, but many pixels on the boundary between positive changed area and unchanged area are still been misclassified. With a robust DI and abundant labeled samples, the classifier of GASCD can be fully trained to achieve a better CM. As reported in Table VII, the accuracies of SNR-GKIT and SNR-FLICM are higher than PCANet and NR-ELM, which benefit from the SNR operator to generate a robust different image to describe the difference information more precisely. Among the proposed methods, GASCD produces the highest PCC and KC values of 96.39% and 0.8676, superior to other compared algorithms, indicating the superiority and effectiveness of the proposed GASCD framework.

4) Experiment on Three SAR Datasets for BCD: Considering about the situation that some SAR image pairs contain only one certain changed type, we experiment our GASCD with other compared algorithms on three datasets for BCD, where the Coastline dataset and the Ottawa dataset only exist positive changes, and the Farmland dataset only contains negative changes. For convenient comparison, we uniformly use the white color to represent the changed area and use black color to represent unchanged area. Apart form the above eight algorithms, another popular BCD method CWNN [2] is introduced for comparison. The CMs and quantitatively

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evaluation values with all competitors are shown in Fig. 20 and Table VIII, respectively.

On the Farmland dataset (the first row of Fig. 20), we can observe that the CM of SNR + FLICM wrongly detected many unchanged regions as changed, which is afflicted with high FP value. CMs of SNR + GKIT and MOGMM tend to be rather noisy. With the least noise, PCANet made a good performance on the Farmland dataset. But on the Coastline dataset, PCNet, NR-ELM, and CWNN are unable to effectively identify the changes from the background noise, which result in higher FPs and lower accuracies. By contrast, the CM generated by GASCD outperforms other methods visually and quantitatively. In addition, on the Sulzberger dataset shown in the third row of Fig. 20, CMs of SNR + GKIT, SNR + FLICM, and NR-ELM still could not fully detect the entire changed area resulting in high FP values. CWNN and MOGMM mistakenly identified many unchanged areas as changes, which leading to the high FN and OE values. Among these nine algorithms, only SAE + CNN, MsCapsNet, and GASCD made better change results on all of the three datasets, indicating that the three methods effectively utilized DNN models to learn the binary difference representation. Compared with SAE + CNN and MsCapsNet, our GASCD still achieves the best OE, PCC, and KC values on each SAR image dataset, which proves that the proposed GASCD method also has a better CD performance on the BCD task.

VI. Conclusion

In this article, we propose an M-nary SAR image CD framework GASCD, which identifies the number of change classes in the temporal image pair and further classifies the changes into positive and negative changes. The proposed GASCD benefits from two aspects; first, to generate a robust DI, an SNR operator has been designed to highlight the salient changes while suppressing the irrelevant noise information. Next, to solve the problem of insufficient samples, a GAN architecture search-based sample generation strategy is applied to produce reliable labeled samples of both positive and negative changed categories. During the labeled sample generation, we decouple the GAN architecture search process into two phases so as to improve the stability of GAN training. Visual experimental results and quantitative analysis show the efficiency and superiority of our GASCD on both TCD and BCD tasks. In the future, we will do research on the sample scarcity dilemma of multiclassification CD on hyperspectral RS images.

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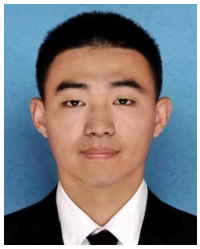

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